

CSE 256 | Lecture 12: Pretrained Decoders

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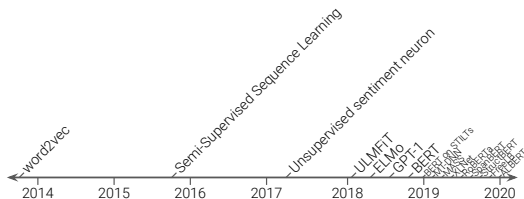
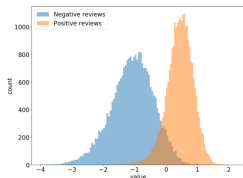
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Today

- ① Decoder LMs (GPT, ...)
- ② Decoding Methods
- ③ Encoder-Decoder LMs (BART, T5)

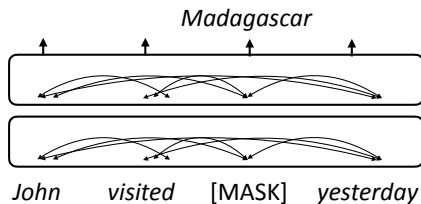
Recap: Pretraining; Learning Representations

- ▶ **Pretraining:** learn useful representations from unlabeled data, then fine-tune for specific tasks with labeled data
- ▶ 2017: Unsupervised Sentiment Neuron
- ▶ 2018: ELMo, BERT, GPT-1, ...



Recap: BERT Objective

- ▶ BERT objective: *masked language modeling*
- ▶ Best version of this: **DeBERTa**, good at NLI/QA/classification tasks



Optimize: $P(\text{Madagascar} \mid \text{John visited [MASK] yesterday})$

Decoders

Decoder Language Models: GPT-2

Radford et al., 2019 : Language Models are Unsupervised Multitask Learners

- ▶ Transformer **decoder** architecture
 - ◉ **Unidirectional (causal) self-attention**, each token attends only to previous tokens
- ▶ **GPT-2 (2019)**
 - ◉ 1.5B parameters
 - ◉ 12 layers, 12 attention heads
 - ◉ hidden size 768
- ▶ Trained with the **autoregressive language modeling objective**

$$P(x_{1:T}) = \prod_{i=1}^T P(x_i \mid x_{<i})$$

GPT-2: Toward Generalized Transfer Learning

Radford et al., 2019 : Language Models are Unsupervised Multitask Learners

- ▶ Before GPT-2: **Pretraining + supervised fine-tuning** e.g. ELMo, BERT
 - ◉ Each task still **required labeled data for fine-tuning**
- ▶ GPT-2 asked a different question:
 - ◉ Do large language models **already contain the capabilities we need?**
- ▶ Key insight: a sufficiently large LM can perform many tasks
 - ◉ **Zero-shot** → without additional training.
 - ◉ By conditioning on a task description in the prompt

Why GPT-2 Can Do Translation (Without Fine-Tuning)

- ▶ GPT-2 was trained on large-scale web text (WebText)
- ▶ The web naturally contains: translations, question–answer pairs, summaries
- ▶ These act as **implicit task demonstrations**
 - ◉ The model learns to continue patterns it has already seen.

”I’m not the cleverest man in the world, but like they say in French: **Je ne suis pas un imbecile** [I’m not a fool].

In a now-deleted post from Aug. 16, Soheil Eid, Tory candidate in the riding of Joliette, wrote in French: **”Mentez mentez, il en restera toujours quelque chose,”** which translates as, **”Lie lie and something will always remain.”**

“I hate the word ‘**perfume**,’” Burr says. ‘It’s somewhat better in French: ‘**parfum**.’

If listened carefully at 29:55, a conversation can be heard between two guys in French: **“-Comment on fait pour aller**

GPT-2: Zero-Shot Task Performance

- **Key idea:** Treat downstream tasks as **language modeling problems**
 - No task-specific training
 - No parameter updates
 - Only prompting / conditioning

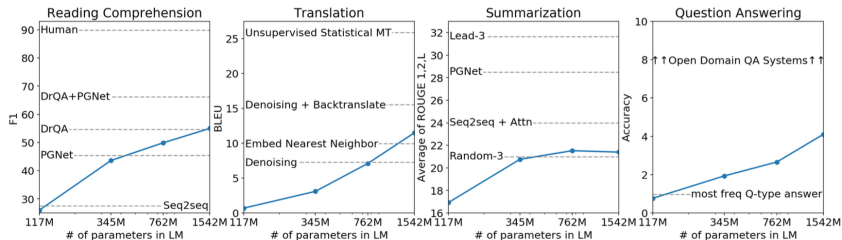


Figure 1. Zero-shot task performance of WebText LMs as a function of model size on many NLP tasks. Reading Comprehension results are on CoQA (Reddy et al., 2018), translation on WMT-14 Fr-En (Artetxe et al., 2017), summarization on CNN and Daily Mail (See et al., 2017), and Question Answering on Natural Questions (Kwiatkowski et al., 2019). Section 3 contains detailed descriptions of each result.

GPT-2: Zero-Shot Summarization

- ▶ The web frequently contains the pattern:
 - ⊙ Article
 - ⊙ TL;DR: short summary
- ▶ GPT-2 learns this pattern during pretraining. At test time:
 - ⊙ Provide an article; Append TL;DR:; Let the model continue

	R-1	R-2	R-L	R-AVG
Bottom-Up Sum	41.22	18.68	38.34	32.75
Lede-3	40.38	17.66	36.62	31.55
Seq2Seq + Attn	31.33	11.81	28.83	23.99
GPT-2 TL;DR:	29.34	8.27	26.58	21.40
Random-3	28.78	8.63	25.52	20.98
GPT-2 no hint	21.58	4.03	19.47	15.03

Table 4. Summarization performance as measured by ROUGE F1 metrics on the CNN and Daily Mail dataset. Bottom-Up Sum is the SOTA model from (Gehrmann et al., 2018)

Decoding Strategies

A language model gives us probabilities.
But it does not give us text.

Decoding Strategies

How Do We Actually Generate Text from a Language Model?

- ▶ A decoder LM defines: $P(x_i \mid x_{<i})$
 - ◉ To generate text, we must **choose** a token at each step.
- ▶ **Greedy decoding**
 - ◉ Choose the most probable token at each step
 - ◉ **Problem:** may get stuck in a loop, or choose a suboptimal path
- ▶ **Beam search**
 - ◉ Approximate the most probable full sequence
 - ◉ Keep track of the top k most probable partial generations at each step
- ▶ **Sampling**
 - ◉ Draw tokens from the probability distribution

Problems with Beam Search

- ▶ Beam search tries to **maximize sequence probability**
- ▶ But maximum likelihood $\neq$ high-quality text
- ▶ For long sequences, maximization leads to:
 - ◉ generic phrasing
 - ◉ repetition
 - ◉ unnatural continuations

When Is Beam Search Appropriate?

It depends on the task

Directed vs. Open-Ended Generation

Holtzman et al., 2019: *The Curious Case of Neural Text Degeneration*

► Open-ended generation

- ◉ Many valid continuations
- ◉ Goal: human-like diversity and creativity
- ◉ Beam search often degenerates

► Directed generation

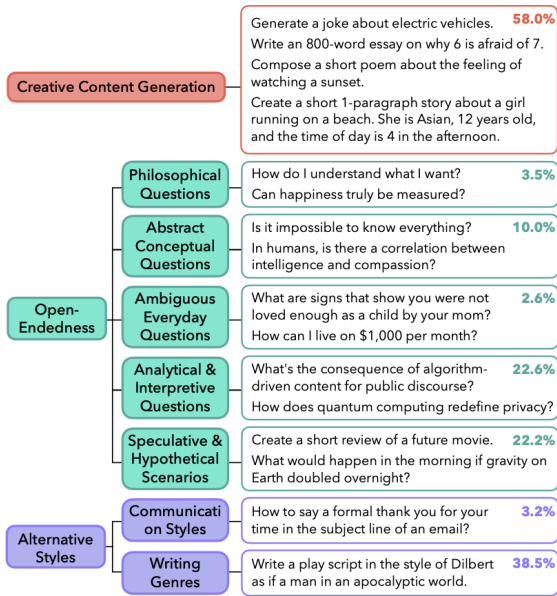
- ◉ Output is constrained by an input
- ◉ Tasks have structured (input, output) pairs
- ◉ Examples:
 - Machine translation
 - Summarization
 - Data-to-text

► In directed settings:

- ◉ Beam search works well
- ◉ Maximizing likelihood aligns with task goals

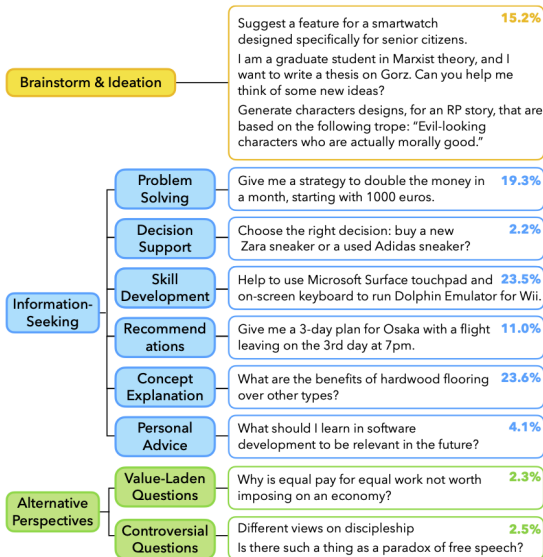
Open-Ended Text Generation

Jiang et al, NeurIPS 2025: *The Open-Ended Homogeneity of LMs*



Open-Ended Text Generation

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If maximizing is too rigid ...
what about sampling?

How Do We Sample a Token?

At time step i , the model outputs scores for each token in the vocabulary:

$$u_1, u_2, \dots, u_{|V|}$$

Convert to probabilities via softmax:

$$p_j = \frac{\exp(u_j)}{\sum_k \exp(u_k)}$$

Now we must sample a token from:

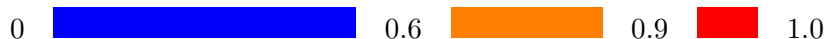
$$\text{Categorical}(p_1, \dots, p_{|V|})$$

Sampling from a Categorical Distribution

Suppose the model predicts: $p = (0.6, 0.3, 0.1)$ for a 3 tokens

Cumulative distribution: $(0.6, 0.9, 1.0)$

Unit Interval Interpretation



Token 1 (60%)

Token 2 (30%)

Token 3 (10%)

- ▶ Draw $r \sim \text{Uniform}(0, 1)$,
 - ⊙ pick the token whose interval contains r
 - ⊙ If $r = 0.45$, we pick token 1; if $r = 0.75$, we pick token 2, ...
- ▶ Token 1 occupies 60% of the interval. Therefore, 60% of uniform draws fall there.

Method: Pure Sampling

Pure Sampling:

They were cattle called **Bolivian Cavalleros**; they live in a remote desert **uninterrupted by town**, and they speak **huge, beautiful, paradisiacal Bolivian linguistic thing**. They say, 'Lunch, marge.' They don't tell what the lunch is," director Professor Chuperas Omwell told Sky News. "They've only been talking to scientists, like we're being interviewed by TV reporters. We don't even stick around to be interviewed by TV reporters. Maybe that's how they figured out that they're cosplaying as the Bolivian Cavalleros."

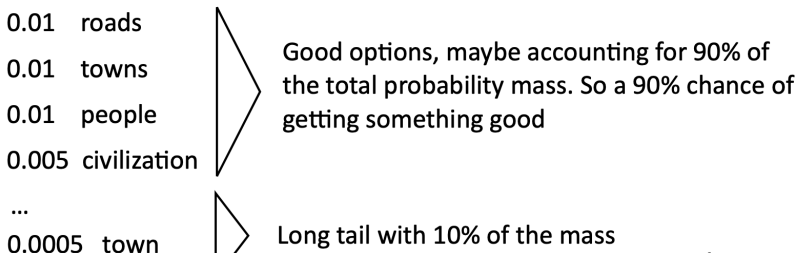
- ▶ **Issue:** the probability distribution has a long **unreliable tail**
 - ◉ Thousands of **low-probability tokens** remain possible
 - ◉ Sampling occasionally picks these unstable options leading to **incoherent text, topic drift, abrupt changes**

Challenges with Pure Sampling

Pure Sampling:

They were cattle called Bolivian Cavalleros; they live in a remote desert uninterrupted by town and they speak huge, beautiful, paradisiacal Bolivian linguistic thing. They say, 'Lunch, marge.' They don't tell what the lunch is," director Professor Chuperas Omwell told Sky News. "They've only been talking to scientists, like we're being interviewed by TV

$P(y \mid \dots \text{they live in a remote desert uninterrupted by})$



Method: Nucleus Sampling (Top- p)

- ▶ Idea: truncate the **unreliable tail** of the distribution
- ▶ Given $P(x \mid x_{<i})$ define the **nucleus** $V^{(p)}$ as the smallest set of tokens such that

$$\sum_{x \in V^{(p)}} P(x \mid x_{<i}) \geq p$$

- ▶ Only tokens in $V^{(p)}$ remain eligible for sampling

Renormalization:

$$P'(x \mid x_{<i}) = \begin{cases} \frac{P(x|x_{<i})}{p'} & x \in V^{(p)} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{where } p' = \sum_{x \in V^{(p)}} P(x \mid x_{<i})$$

Nucleus Sampling: Intuition and Implementation

$P(y \mid \dots \text{they live in a remote desert uninterrupted by})$

0.01 roads

0.01 towns

0.01 people

0.005 civilization



renormalize and sample

cut off after $p\%$ of mass

- ▶ Keep only the smallest set of tokens, whose total probability mass $\geq p$, Drop the unreliable tail
- ▶ **Implementation:**
 - ◉ Sort options by probability, truncate the list once the total exceeds p
 - ◉ Renormalize, Sample

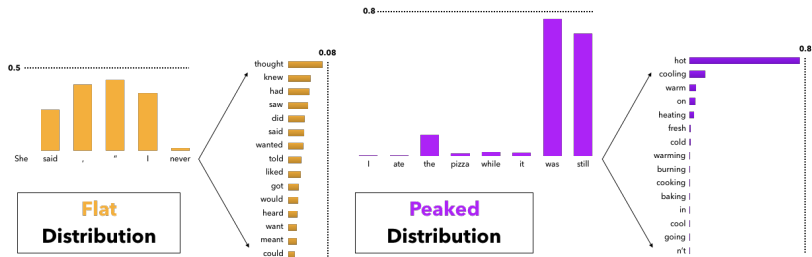
Another Way to Truncate the Tail

Instead of fixing probability mass (p),
what if we fix the number of tokens?

Method: Top- k Sampling

- ▶ At each time step:
 - ◉ Keep only the k highest-probability tokens
 - ◉ Renormalize, sample
- ▶ Difference from nucleus sampling:
 - ◉ **Top- k** : fixed vocabulary size
 - ◉ **Top- p** : fixed probability mass
- ▶ Both restrict sampling to a *trustworthy region*

Top- k Sampling: Challenges of a Fixed k



- If the distribution is **flat**:
 - ◉ A small k may discard many plausible tokens
 - ◉ Sampling becomes overly restrictive
- If the distribution is **peaked**:
 - ◉ A large k includes many low-quality tail tokens
 - ◉ Sampling becomes unstable

A fixed k ignores distribution shape.

Top- p vs Top- k : When to Use Which?

Top- p (Nucleus Sampling)

- ▶ Adapts to the shape of the distribution
- ▶ Works well when entropy varies across steps

Why it became popular:

- ▶ Produces more human-like diversity

Top- k Sampling

- ▶ Keeps exactly k highest-probability tokens
- ▶ Simpler and computationally cheaper

When to use it:

- ▶ Low-entropy, predictable domains (e.g., code generation, constrained tasks)

Another Way to Control Randomness

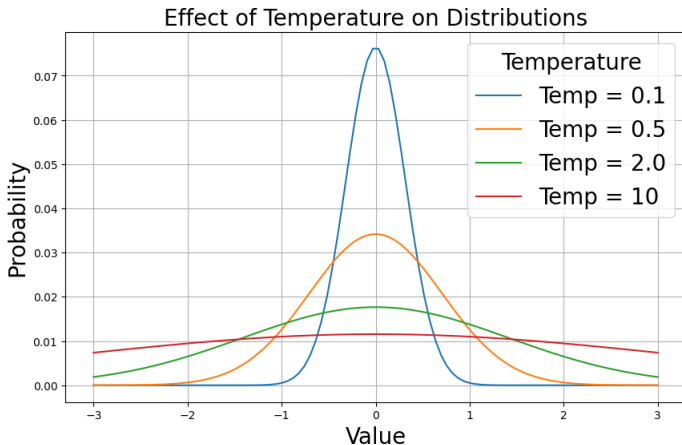
Instead of removing tokens,
what if we reshape the distribution itself?

Sampling with Temperature

High temperature: more randomness; **Low temperature:** less randomness

Given the scores $u_{1:|V|}$ and temperature t , adjust softmax as:

$$p(x = V_l \mid x_{1:i-1}) = \frac{\exp(u_l/t)}{\sum_{l'} \exp(u_{l'}/t)}$$



Temperature: Numerical Example

Unnormalized scores:

$$u = [2, 1, 0]$$

Original ($t = 1$): $[0.665, 0.245, 0.090]$

Low temperature ($t = 0.5$): $[0.867, 0.117, 0.016]$

High temperature ($t = 2$): $[0.506, 0.307, 0.186]$

Really High temperature ($t = 10$): $[0.337, 0.332, 0.331]$

Ranking unchanged. Confidence changes.

Temperature: Trade-offs

- ▶ Low temperature:
 - ◉ concentrates mass on high-probability tokens
 - ◉ improves coherence
 - ◉ But reduces diversity (Caccia et al., 2018; Hashimoto et al., 2019).
- ▶ High temperature:
 - ◉ increases diversity
 - ◉ increases risk of incoherence
- ▶ Temperature reshapes the distribution
- ▶ It does **not** remove tokens

How Do We Evaluate Decoding?

Does higher probability mean better text?

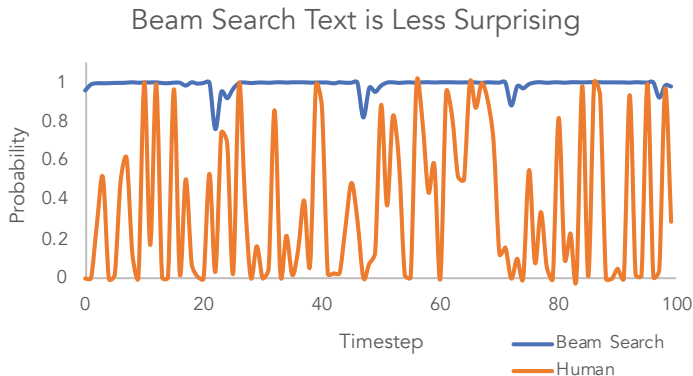
Decoding Strategies: Perplexity

- ▶ Beam search often achieves lower perplexity
- ▶ But lower perplexity does not imply more human-like text

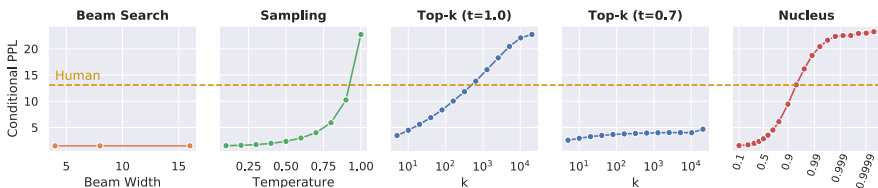
Method	Perplexity
Human	12.38
Greedy	1.50
Beam, $b=16$	1.48
Stochastic Beam, $b=16$	19.20
Pure Sampling	22.73
Sampling, $t=0.9$	10.25
Top- $k=40$	6.88
Top- $k=640$	13.82
Top- $k=40$, $t=0.7$	3.48
Nucleus $p=0.95$	13.13

Natural Language Does Not Maximize Probability

- ▶ **Observation:** Natural language text, as written by humans, does not maximize per-token probability
- ▶ Natural language incorporates lower-probability but informative tokens



Perplexities of Generations from Various Decoding Methods



- ▶ **Beam Search:** Shows unnaturally low perplexities, indicating overly predictable sequences.
- ▶ **Sampling, Top- k , and Nucleus** Sampling can be adjusted to achieve human-like perplexities.
- ▶ However, **Sampling and Top- k** struggle with coherency when parameters are tuned to reach these human-level perplexities.

Decoding Strategies: The Design Space

- ▶ A language model defines: $P(x_i \mid x_{<i})$
- ▶ Decoding is a **policy** for choosing from this distribution.
- ▶ **Maximization-based decoding**
 - ◉ Greedy, Beam
 - ◉ Optimize likelihood
 - ◉ Risk: repetition, generic text
- ▶ **Sampling-based decoding**
 - ◉ Pure sampling, Top- k , Top- p , Temperature
 - ◉ Control randomness
 - ◉ Tradeoff: coherence vs diversity

Decoding balances probability and surprise.

Encoder-Decoder Models

Combining Encoders and Decoders for Pretraining

- ▶ So far:
 - ◉ Encoders model representations of $\mathbf{x}$
 - ◉ Decoders model $P(\mathbf{x})$ via next-token prediction
- ▶ But many tasks require:

$$P(\mathbf{y} \mid \mathbf{x})$$

- ◉ Translation
- ◉ Summarization
- ◉ Question answering

Read with full context. Write left-to-right.

How Do We Pretrain Encoder–Decoder Models?

- ▶ We want to learn:

$$P(\mathbf{y} \mid \mathbf{x})$$

without labeled $(\mathbf{x}, \mathbf{y})$ pairs.

- ▶ Requirements:

- ◉ Use only raw text (self-supervised)
- ◉ Force the decoder to attend to encoder representations

- ▶ Key idea: **Create synthetic input–output pairs from text itself**

BART Pretraining: Denoising Objective

- ▶ Start with a clean sentence:

“The cat sat on the mat.”

- ▶ Apply a corruption function to create input $\tilde{\mathbf{x}}$
- ▶ Train encoder–decoder to reconstruct the original:

$$P(\mathbf{x} \mid \tilde{\mathbf{x}})$$

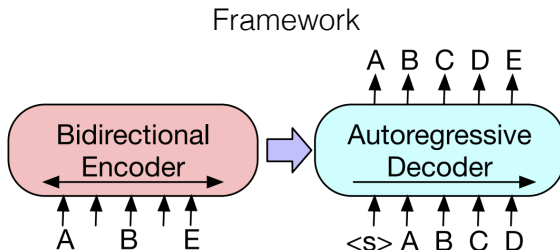
Corruption strategies explored in BART:

- ▶ Token masking (like BERT)
- ▶ Token deletion
- ▶ Text infilling (span masking)
- ▶ Sentence permutation

Learn to reconstruct meaning from noisy input.

BART Pretraining

- Encoder-Decoder Transformer trained on this data: (permute/mask/delete tokens), then predict full sequence autoregressively



BART Applied to Text Summarization

- ▶ **Pre-train** BART: Corrupt random text segments and decode the original text.
- ▶ **Fine-tune** on text summarization: Use news articles as input to generate brief summaries (1-3 sentences).
- ▶ **Few summaries needed for fine-tuning**: Achieves good results with fewer examples compared to traditional seq2seq models, which need over 10,000 examples.

BART for Summarization: Outputs

PG&E stated it scheduled the blackouts in response to forecasts for high winds amid dry conditions. The aim is to reduce the risk of wildfires. Nearly 800 thousand customers were scheduled to be affected by the shutoffs which were expected to last through at least midday tomorrow.

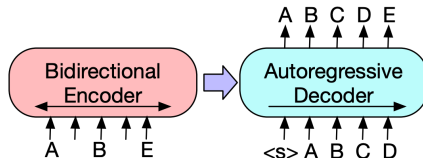
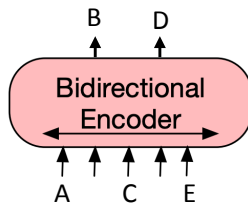


Power has been turned off to millions of customers in California as part of a power shutoff plan.

Lewis et al. (2019)

BERT vs. BART

- ▶ **BERT**: bidirectional encoder only, masked LM
- ▶ Cannot generate text or do seq2seq tasks such as translation
- ▶ **BART**: bidirectional, autoregressive
- ▶ Both an encoder and a decoder. Can also use just the encoder wherever we would use BERT



T5

Text-to-Text Transfer Transformer

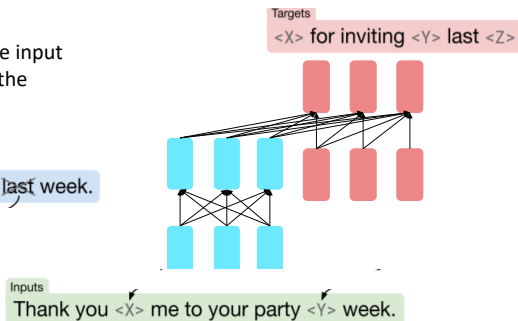
- ▶ Treat all tasks as text-to-text problems
- ▶ Pre-training: similar denoising scheme to BART (they were released within a week of each other in fall 2019)
- ▶ Input: text with gaps; Output: a series of phrases to fill those gaps

Replace different-length spans from the input with unique placeholders; decode out the spans that were removed!

Original text

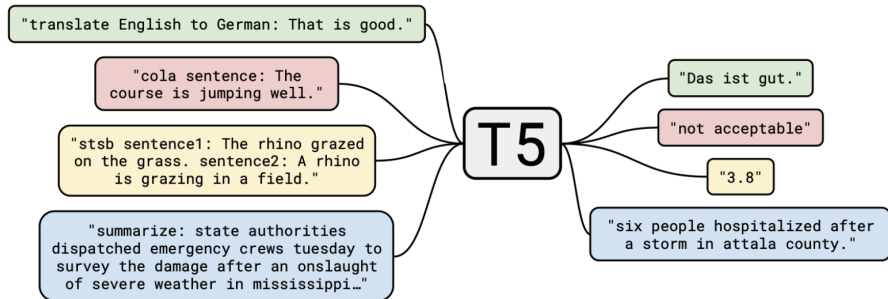
Thank you for inviting me to your party last week.

This is implemented in text preprocessing: it's still an objective that looks like **language modeling** at the decoder side.



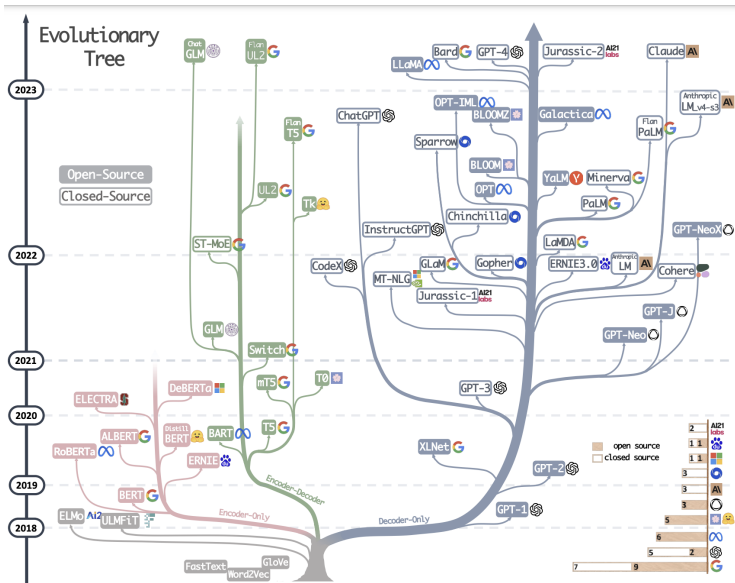
T5: Supervised Multitask Learning

- ▶ After **span corruption pretraining**, continue training with **supervised multitask learning**
 - ◉ Formulate tasks as text-to-text format
 - ◉ Use a prefix to denote the task
 - ◉ Mixing examples from different datasets when constructing batches



Pretraining: Transformer Architectures

Pretraining: encoder, decoder, encoder-decoder, which to use?



Why Decoder-Only Models Dominate Modern LLMs

- ▶ **Unified Objective:** Same objective at every stage
 - ◉ Next-token prediction during:
 - ▶ Pretraining
 - ▶ Post-training
 - ▶ Prompting (zero-shot / few-shot)
- ▶ **Generality:** Can perform a wide range of tasks
 - ◉ Naturally generative (chat, code, storytelling)
 - ◉ Can perform classification via prompting or fine-tuning
- ▶ **Simplicity**
 - ◉ No masking or corruption strategies required

What Language Model Architecture and Pretraining Objective Work Best for Zero-Shot Generalization?

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Julien Launay^{3,5†} Colin Raffel^{1†}

Abstract

Large pretrained Transformer language models have been shown to exhibit *zero-shot generalization*, i.e. they can perform a wide variety of tasks that they were not explicitly trained on. However, the architectures and pretraining objectives used across state-of-the-art models differ significantly, and there has been limited systematic comparison of these factors. In this work, we present a large-scale evaluation of modeling choices and their impact on zero-shot generalization. In particular, we focus on text-to-text models and experiment with three model architectures (causal/lstm, causal/decoder

1. Introduction

Large language models (LLMs) pretrained on unstructured text data have been shown to be capable of performing a wide variety of text processing tasks without additional training or labeled data. This ability has been referred to as **zero-shot generalization** since these models are typically pretrained with a self-supervised objective with no regards to specific downstream tasks. As such, there has been an explosion of work on developing LLMs and training techniques that produce strong zero-shot generalization (Brown et al., 2020; Wang and Komatsuzaki, 2021; Du et al., 2021; Lin et al., 2021; Rae et al., 2021; Hoffmann et al., 2022; Chowdhery et al., 2022).

Finding 1. **Causal decoder-only** models pretrained with a **full language modeling** objective achieve best zero-shot generalization when evaluated immediately after **self-supervised pretraining**, in line with current common practices for large language models.

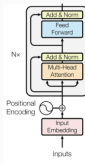
Questions?

Decoder-only GPT



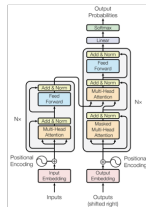
Encoder-only BERT

[*] [*] [sat_] [*] [the_] [*]



Enc-Dec T5

Das ist gut.
A storm in Attala caused 6 victims.
This is not toxic.



Translate EN-DE: This is good.
Summarize: state authorities dispatched..
Is this toxic: You look beautiful today!

Next Lecture: Scaling Laws and Efficient Adaptation